development of string theory, finding its “principle of inevitability”—that underlying idea from which the whole theory necessarily springs forth—is of the highest priority.2

What Are Space and Time, Really, and Can We Do without Them?

In many of the preceding chapters, we have freely made use of the concepts of space and of spacetime. In Chapter 2 we described Einstein’s realization that space and time are inextricably interwoven by the unexpected fact that an object’s motion through space has an influence on its passage through time. In Chapter 3, we deepened our understanding of spacetime’s role in the unfolding of the cosmos through general relativity, which shows that the detailed shape of the spacetime fabric communicates the force of gravity from one place to another. The violent quantum undulations in the microscopic structure of the fabric, as discussed in Chapters 4 and 5, established the need for a new theory, leading us to string theory. And finally, in a number of the chapters that followed, we have seen that string theory proclaims that the universe has many more dimensions than we are aware of, some of which are curled up into tiny but complicated shapes that can undergo wondrous transformations in which their fabric punctures, tears, and then repairs itself.

Through graphic representations such as Figures 3.4, 3.6, and 8.10, we have tried to illustrate these ideas by envisioning the fabric of space and spacetime as if it were somewhat like a piece of material out of which the universe is tailored. These images have considerable explanatory power; they are used regularly by physicists as a visual guide in their own technical work. Although staring at figures such as the ones just mentioned gives a gradual impression of meaning, one can still be left asking, What do we really mean by the fabric of the universe?

This is a profound question that has, in one form or another, been the subject of debate for hundreds of years. Newton declared space and time to be eternal and immutable ingredients in the makeup of the cosmos, pristine structures lying beyond the bounds of question and explanation. As he wrote in the Principia, “Absolute space, in its own nature, without relation to anything external, remains always similar and immovable. Absolute, true, and mathematical time, of itself, and from its own nature, flows equably without relation to anything